

YAMAKAWA-ZHANDRY 2022 : QUANTUM ADVANTAGE WITHOUT STRUCTURE

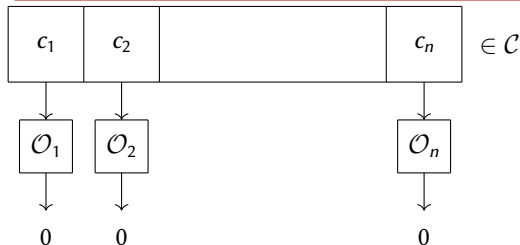
Defined a problem that is hard classically, but easy quantumly, related to a random oracle only.

Let $\mathcal{C}$ be a linear subspace of $\mathbb{F}_q^n$ (linear code).

Given n random oracles $\mathcal{O}_i : \mathbb{F}_q^n \rightarrow \{0, 1\}$.

Find $c \in \mathcal{C}$ s.t. $\mathcal{O}_i(c_i) = 0$ for all $i \in [n]$.

$\mathcal{O}_i(x)$ is truly random until x is queried, then it's fixed.



Classical hardness comes from **global** + **local** constraints.

Easy quantumly:

- “short vector”: $\mathbf{x}$ s.t. $\mathcal{O}_i(\mathbf{x}_i) = 0$ for all $i \longrightarrow \hat{f}(\mathbf{x}) = 1$ if $\mathcal{O}_i(\mathbf{x}_i) = 0$ for all i ;
- choose an appropriate code $\mathcal{C}$ so that f is efficiently decodable.